Testing for Significance and Multicollinearity Transcript

Then we will learn how to test our model significance. So we will learn about multicollinearity and then we will learn about variance inflation factor and then we will also try to see how do we use F test for understanding whether the model that we have created with all of these input variables is actually significant or not. So to test for model significance we use F test, to test for variable significance we use T test and then P values.

To test whether a set of variables that you have included are they actually affecting the target variable, so we use a metric called as adjusted R square which basically assesses the amount of variance which is there in the output variable. Well how much of that is being explained by the input variables that you have captured or that you have added in the model. So with that let's get started and let's get introduced to lesson 2. Hey hi everyone, now how do we test for significance of the model and how do we test with the model that we are using, are they actually good or not, whether the predictors or the coefficient of the predictors are significant, do they overlap, I mean we will look that into multicollinearity aspects and then so on.

But the main question is how do we say whether the model is useful or not, how do we say whether the predictors are significant or not, this is what we are going to learn in the session. First we have two levels of significance testing, one overall level which is at the entire model level, the second at the individual predictor level, so each predictor, each input variable, for example you have these predictors right, swimming pool, bedroom, square foot, so for each predictor we do individual tests, whether this predictor is actually contributing to the model as in whether swimming pool is affecting price or not, whether bedroom price is actually affecting price or not, whether square foot is actually affecting price or not. So in total we first do a test for the entire model, all data taken together and then seeing whether price is getting affected because of all of them okay, and then once we know that the model is good, we test for individuals, we test whether swimming pool individually is affecting price or not okay, that's what we are learning here in this entire significance testing and multicollinearity lecture.

Our idea is to test in two ways, one for the overall model, we will try to see if the overall F statistic is significant or let's say for example F statistic value is making sense or not, then we will look into individual t-test values for each predictors and see if those are also making sense or not okay. So this is the normal criteria that we use right, we have null hypothesis which says that all the individual values or the parameters they are 0, otherwise they are at least one of them is not 0, so when I say at least one of them is not 0, it means that there could be one variable which could affect Y out of hundreds or thousands that you have accumulated data for, but there could be multiples as well right, there could be one, there could be two, there could be some combination, idea is you frame the hypothesis and if the p-value attached to F statistics is less than 0.05, then you say that the model is significant, it means at least one of those variables, input variables is affecting price okay, the example that I gave you. Now once we know that the model is meaningful, once we know that the model is significant, then we go deeper and we try testing which particular parameters or which particular variables are actually affecting the target variable, so then we go ahead and decide whether swimming pool is actually affecting price or bedrooms is affecting price or square foot is affecting price, remember this is one simple example that I have taken, you could have any example when you start working on real life data okay, so then what we say for that particular variable, let's say swimming pool in our example, if its value is equals to 0 or the value of the coefficient is equals to 0 or the value of the coefficient is not equals to 0, because these are the values which are attached to the variables right, beta j into xj, xj is the value of the variable and beta j is nothing but the coefficient of that, if the coefficient is 0, it means that the variable is not making any sense or it is not affecting our target variable or y variable right.

Now for doing this we use individual key test, if the p value for those test is significant which is less than 0.05, we say that the alternate hypothesis is true which means that variable is actually affecting the target variable okay, what happens when we have two variables which are input or two input variables which are interdependent on each other or they are dependent on each other right, that situation is called as multicollinearity, multicollinearity affects when one variable is or let's say let me reframe that sentence, it affects when two or more predictor variables are actually dependent on each other, they are correlated with each other okay, that situation is called as multicollinearity, imagine you are trying to model a person's fitness level using time spent on running and the distance run right, these two variables are directly correlated, the distance and time are correlated with each other right, so in that case there are implications to the model and there are various different types of problems that you can have when your input variables in the case of multiple linear regression, when your input variables are correlated with each other then there are a lot of problems that the model has to go through, so we will understand all of those things here, first thing is if there is multicollinearity which exist in the data then you could have wrong signs of the coefficient okay, then the coefficient might also become very much unstable, what does that mean? It means that if the real value of the coefficient is 2 they could become highly inflated, they could assume a value which is 2000 or 20000 or very high value, in that case you will end up saying that the variable is actually affecting the y variable right, so they become inflated when there is multicollinearity in the data okay, the overall F-test is significant, it would come down as significant but the individual t-test for the correlated predictors are not, so this is also one of the symptoms using which you can actually sense whether there is multicollinearity or not, but very simple ways of understanding whether there is multicollinearity, these are two problems that I have just told you right, but to understand whether there is multicollinearity in your data or not, you can use variance inflation factor, this is just using correlation matrix or let's say you use a simple correlation matrix wherein if the correlation between two variables is more than 0.8, you randomly remove one of them, that's what generally people do but there are other methods as well, the idea is do not keep any variable which is correlated to all other variables because it causes these two problems okay, hence in that case the model would be insignificant because you will be making wrong conclusions about the model okay, so how to remove that if your correlation matrix says two variables are really highly correlated, their correlation between them is greater than 0.8, hence you would be removing one of them randomly, you also use variance inflation factor VIF which means if the value of the VIF let's say, VIF is directly dependent on R square value or adjusted R square value, if VIF is greater than 5 or 10, it means that those variables are problematic and hence it is good to remove those variables, so we calculate VIF for all the variables and for the ones where the VIF value is greater than 5 and 10, those are the ones which we remove, I hope this is clear, this is all clear to everyone.

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